

Unit 6: Wrap-up & Further Resources

Dr Christian Engels
christian_engels@outlook.com

Applied Timeseries in R
7 Feb 2026

Certificate in Engagement with Practice in Finance (CEPF)
Durham University Business School



Workshop Recap

Key R Packages

Further Reading

Next Steps

Workshop Recap

What We Covered Today

Unit	Topic	Key Concepts
0	Setup	GitHub Codespaces, R environment, package installation
1	TS Foundations	Tidyverse workflow, tsibble, log returns, visualisation
2	Random Walks	White noise, stationarity, ADF/KPSS tests, ACF/PACF
3	ARIMA	AR, MA, ARIMA models, model selection, forecasting
4	GARCH	Volatility clustering, sGARCH, GJR-GARCH, eGARCH, VaR
5	Portfolios	Mean-variance optimisation, GARCH on portfolios, backtesting

- ➊ **Data:** Download and prepare financial data (`tidyfinance`, `tidyverse`)
- ➋ **Explore:** Visualise, compute returns, check distributions (`ggplot2`, `tsibble`)
- ➌ **Test:** Stationarity tests, ACF/PACF analysis (`feasts`, `tseries`)
- ➍ **Model:** Fit ARIMA (mean) and GARCH (variance) (`fable`, `rugarch`)
- ➎ **Diagnose:** Residual analysis, model comparison (AIC/BIC, Ljung–Box)
- ➏ **Forecast:** Out-of-sample predictions, rolling forecasts
- ➐ **Apply:** Portfolio optimisation, VaR, backtesting (`PortfolioAnalytics`)

Key R Packages

Package	Purpose
<code>tidyverse</code>	Data manipulation, visualisation, piping
<code>tidyfinance</code>	Download financial and economic data
<code>tsibble</code>	Time series data structure for tidy workflows
<code>fable</code>	Time series modelling (ARIMA, ETS, etc.)
<code>feasts</code>	Features extraction, ACF/PACF, decomposition
<code>rugarch</code>	GARCH specification, fitting, forecasting, VaR
<code>PortfolioAnalytics</code>	Portfolio optimisation with constraints
<code>xts / zoo</code>	Time series objects for rugarch
<code>tseries</code>	ADF test, KPSS test, Jarque–Bera test
<code>scales</code>	Axis formatting (percent, dollar labels)
<code>moments</code>	Skewness and kurtosis

Data & Exploration:

`download_data()` — fetch prices
`mutate(ret = log(close/lag(close)))`
`as_tsibble(index = date)`
`ACF()` / `PACF()` — model ID
`adf.test()` / `kpss.test()`
`unitroot_ndiffs()` — diff order

Modelling & Diagnostics:

`model(ARIMA(ret))` — fit ARIMA
`gg_tsresiduals()` — diagnostics
`ugarchspec()` — define GARCH
`ugarchfit()` — estimate GARCH
`infocriteria()` — AIC/BIC
`ugarchroll()` — rolling backtest

Further Reading

Time Series & Forecasting:

- Hyndman & Athanasopoulos, *Forecasting: Principles and Practice* (3e)
Free: <https://otexts.com/fpp3/>
- Hamilton, *Time Series Analysis*
- Tsay, *Analysis of Financial Time Series* (3e)

Finance & Risk:

- Christoffersen, *Elements of Financial Risk Management* (2e)
- McNeil, Frey & Embrechts, *Quantitative Risk Management*
- Ait-Sahalia & Hansen, *Handbook of Financial Econometrics*

- **Tidy Finance with R:** <https://www.tidy-finance.org/r/>
Comprehensive guide to financial data analysis in R
- **R for Data Science (2e):** <https://r4ds.hadley.nz/>
The definitive tidyverse reference
- **CRAN Task View — Time Series:** <https://cran.r-project.org/web/views/TimeSeries.html>
- **CRAN Task View — Finance:** <https://cran.r-project.org/web/views/Finance.html>
- **FRED:** <https://fred.stlouisfed.org>
Federal Reserve Economic Data — free macroeconomic data

Next Steps

Extend Your Skills:

- Multivariate GARCH (DCC, BEKK)
- Cointegration & VECM
- Bayesian time series methods
- Machine learning for forecasting
- High-frequency data analysis

Practice:

- Apply today's methods to your own data
- Reproduce published results
- Contribute to open-source R packages
- Join the R community
R-bloggers, Stack Overflow, Posit Community

Key Takeaways

- Financial returns exhibit **stylised facts**: fat tails, volatility clustering, leverage effects
- **Stationarity** is essential—always test with ADF and KPSS before modelling
- Use **ACF/PACF** patterns to identify model orders; validate with residual diagnostics (Ljung–Box)
- **ARIMA** models the conditional mean; **GARCH** models the conditional variance
- **Diversification** reduces portfolio risk; mean–variance optimisation finds efficient allocations
- **Backtesting** validates risk models—use Kupiec and Christoffersen coverage tests
- The **tidyverse + fable/feasts + rugarch** ecosystem provides a complete financial time series toolkit

Thank You!

Questions?

Contact: christian_engels@outlook.com

Materials & companion R scripts:

<https://github.com/christianengels/applied-timeseries-in-r>